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Video SimpleQA: Towards Factuality Evaluation in Large Video Language Models

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https://videosimpleqa.github.io/

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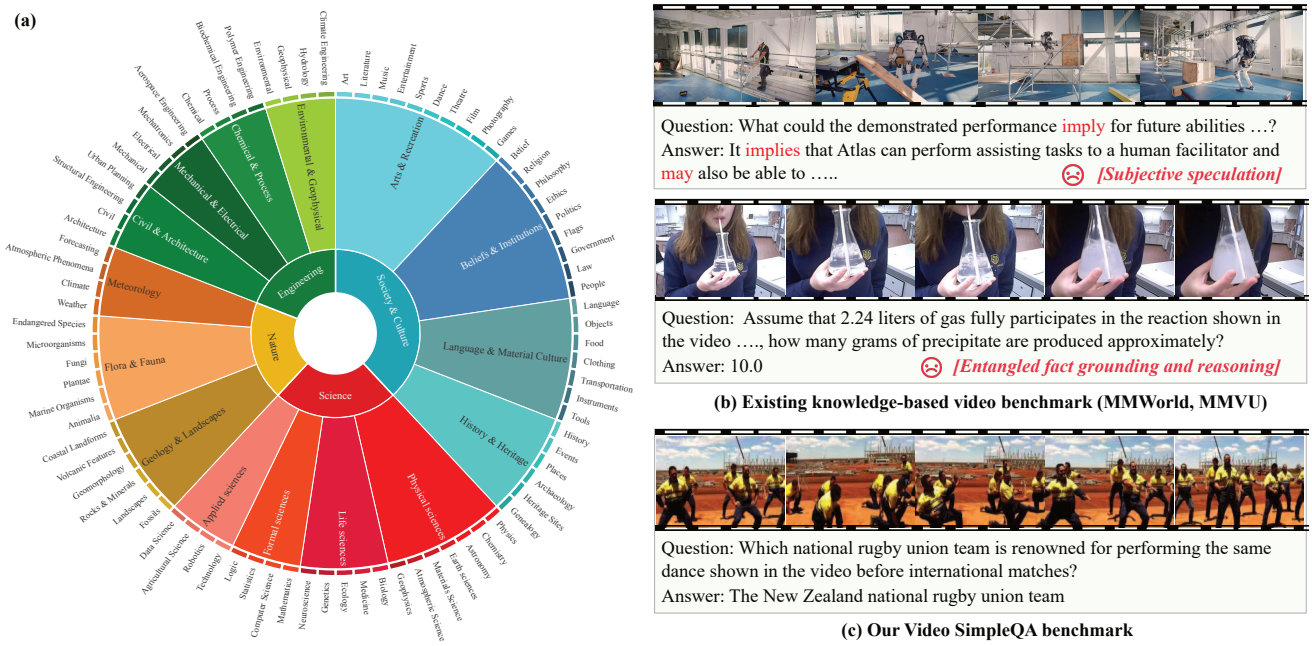


Figure 1: (a) **The taxonomy** of `Video SimpleQA` benchmark; (b) **Illustrations of existing knowledge-based video benchmarks** (Zhao et al. 2025; He et al. 2024a) which may contain subjective speculation or conflate factual grounding with reasoning skills (*i.e.*, mathematical calculation); (c) **Illustrations of our Video SimpleQA benchmark** with the fact-seeking question and definitive & short-form answer with multi-hop external facts verified.

Abstract

Recent advancements in Large Video Language Models (LVLMs) have highlighted their potential for multi-modal understanding, yet evaluating their factual grounding in videos remains a critical unsolved challenge. To address this gap, we introduce `Video SimpleQA`, the first comprehensive benchmark tailored for factuality evaluation in video contexts. Our work differs from existing video benchmarks through the following key features: 1) **Knowledge required**: demanding integration of external knowledge beyond the video's explicit narrative; 2) **Multi-hop fact-seeking question**: Each question involves multiple explicit facts and requires strict factual grounding without hypothetical or subjective inferences. We also include per-hop single-fact-based sub-QAs alongside final QAs to enable fine-grained, step-by-step evaluation; 3) **Short-form definitive answer**: Answers are crafted as unambiguous and definitively correct in a short format with minimal scoring variance; 4) **Temporal grounded required**: Requiring answers to rely on one or more temporal segments in videos, rather than single frames. We extensively evaluate 33 state-of-the-art LVLMs and summarize key findings as follows: 1) Current LVLMs exhibit notable deficiencies in factual adherence, with the best-performing model o3 merely achieving an F-score of 66.3%; 2) Most LVLMs are overconfident in what they generate, with self-stated confidence exceeding actual accuracy; 3) Retrieval-augmented generation demonstrates consistent improvements at the cost of additional inference time overhead; 4) Multi-hop QA demonstrates substantially degraded performance compared to single-hop sub-QAs, with first-hop object/event recognition emerging as the primary bottleneck. We position `Video SimpleQA` as the cornerstone benchmark for video factuality assessment, aiming to steer LVLM development toward verifiable grounding in real-world contexts.



sub-question-1: Which short science-fiction film is excerpted in the video?
sub-answer-1: Tears of Steel
ref. source-1: https://commons.wikimedia.org/wiki/Template:Motd/2013-01-26

sub-question-2: Who is the director of Tears of Steel?
sub-answer-2: Ian Hubert
ref. source-2: https://en.wikipedia.org/wiki/Tears_of_Steel

sub-question-3: Which country is Ian Hubert from?
sub-answer-3: USA
ref. source-3: https://baike.baidu.com/item/Ian%20Hubert

sub-question-4: Who is the first holder of Chief Justice of USA?
sub-answer-4: John Jay
ref. source-4: https://en.wikipedia.org/wiki/John_Jay

multi-hop-question: Who is the first holder of Chief Justice of the country which is the director of the short science-fiction excerted in the video from?
multi-hop-answer: John Jay



sub-question-1: Which country is the tourist attraction in the video located in?
sub-answer-1: Greece
ref. source-1: https://commons.wikimedia.org/wiki/Template:Motd/2011-01-16

sub-question-2: What is the capital of Greece?
sub-answer-2: Athens
ref. source-2: https://en.wikipedia.org/wiki/Greece

sub-question-3: What peninsula is Athens located on?
sub-answer-3: Balkan peninsula
ref. source-3: https://en.wikipedia.org/wiki/Athens

sub-question-4: Who coined the term "Balkan Peninsula"?
sub-answer-4: August Zeune
ref. source-4: https://en.wikipedia.org/wiki/Balkans

multi-hop-question: Who coined the term for the peninsula where the capital of the country containing the tourist attraction in the video is located?
multi-hop-answer: August Zeune

Figure 2: **Four-hop examples** in `Video SimpleQA` including the final multi-hop QA and the decomposed per-fact sub-QAs.

Introduction

The substantial advancements in Large Language Models (LLMs) (Achiam et al. 2023; Reid et al. 2024; Touvron et al. 2023) over the past few years have inaugurated a new frontier in artificial intelligence. Despite their remarkable capabilities, the factuality concern (Wang et al. 2024c; Akhtar et al. 2023; Wang et al. 2023) remains a critical challenge, *i.e.*, how to ensure that the generated contents are consistent with factual knowledge and grounded in credible sources.[1]

Existing research has primarily focused on evaluating factuality in text-based (Yu et al. 2022; Pan et al. 2024; Lin, Hilton, and Evans 2022; Chern et al. 2023; Gou et al. 2023) and image-based (Marino et al. 2019; Wang et al. 2015, 2017; Zellers et al. 2019; Jain et al. 2021) scenarios. Recently, the SimpleQA benchmark (Wei et al. 2024) introduced by OpenAI and its subsequent works (He et al. 2024b; Gu et al. 2025; Cheng et al. 2025) streamline the factuality evaluation by considering only concise and fact-seeking questions, which enables standardized and tractable assessments. However, extending this paradigm to video contexts is under-explored and presents unique challenges due to the inherent temporal dynamics and procedural knowledge.

To bridge this gap, we present `Video SimpleQA`, a comprehensive factuality evaluation benchmark tailored for Large Video Language Models (LVLMs). As shown in Figure 1 and Figure 2, `Video SimpleQA` is composed of multi-hop fact-seeking questions and short-form definitive answers. Compared to previous video benchmarks, `Video SimpleQA` stands out with the following advancements:

- **Knowledge required**: Beyond comprehending the visual content, `Video SimpleQA` necessitates the integration of external knowledge that is not explicitly presented in the video narrative, *e.g.*, domain-specific information, contextual background, commonsense.
- **Multi-hop fact-seeking question**: Questions necessitate strict adherence to factual grounding principles, *eliminating any hypothetical or subjective inferences*. In addition, each question is constructed to involve *multiple explicitly identifiable facts*. To achieve this, beyond the final multi-hop question-answer (QA) pairs, we additionally provide per-fact specific sub-QA annotations, which facilitate fine-grained evaluation of model performance at each fact-grounding hop and help pinpoint exactly which hop fails in factual grounding (*c.f.* Figure 2).
- **Short-form definitive answer**: All the answers are unambiguous, universally agreed upon, consistent over time, and invariant to individual perspectives. Following SimpleQA (Wei et al. 2024), the answers also advocate the short-form paradigm, which establishes reliable factual assessment with low run-to-run variance.
- **Temporal grounded:** Answering questions in `Video SimpleQA` should refer to one or more temporal segments in the video, rather than relying on a single frame.

While existing knowledge-based (Garcia et al. 2020; Zhang et al. 2024b; Hu et al. 2025) and recent discipline-based (Zhao et al. 2025; He et al. 2024a) benchmarks may appear similar to our `Video SimpleQA` , our benchmark features several distinct characteristics (*c.f.* Table 1):

- **Open-domain**: While KnowIT-VQA (Garcia et al. 2020) is constrained to TV shows, and MMVU (Zhao et al. 2025) as well as MMWorld (He et al. 2024a) focus on discipline-specific knowledge, our `Video SimpleQA` encompasses open-domain video types and questions.
- **Objective QA**: Our benchmark is explicitly designed for factuality evaluation through **objective** factual assertions, in contrast to existing benchmarks that often involve varying degrees of subjectivity, even those focusing on disciplinary knowledge. For instance, as shown in Figure 1(b) top, MMWorld (He et al. 2024a) includes cases requiring predictions about a robot's future capabilities—introducing subjective speculation and personal judgment, which deviates from our goal of evidence-based and objective evaluation.[2]
- **Factuality exclusive**: Discipline-based benchmarks of-

[1] Please refer to (Wang et al. 2023, 2024c) for the differentiation between the *factuality* and the similar *hallucination* concepts.

[2] More examples are available in the supplementary material.

Table 1: **Benchmark comparisons** across key dimensions: video domain scope, knowledge-driven focus, objective factuality focus, exclusivity of factual evaluation, multi-hop fact decomposition, and external evidence source.

Benchmarks	Video domain	Knowledge driven	Objective QA	Factuality exclusive	Multi-hop fact decomp.	Evidence source
Video-MME (Fu et al. 2024)	Open	✗	✗	✗	✗	✗
MMBench-Video (Fang et al. 2024)	Open	✗	✗	✗	✗	✗
Video-MMMU (Hu et al. 2025)	Professional	✓	✗	✗	✗	✗
MMVU (Zhao et al. 2025)	Discipline	✓	✗	✗	✗	✓
MMWorld (He et al. 2024a)	Discipline	✓	✗	✗	✗	✗
WorldQA (Zhang et al. 2024b)	Open	✓	✗	✗	✗	✗
KnowIT-VQA (Garcia et al. 2020)	TV shows	✓	✗	✗	✗	✗
`Video SimpleQA`	Open	✓	✓	✓	✓	✓

ten conflate external knowledge retrieval with reasoning skills (*e.g.*, numerical calculations). For example, the case in Figure 1(b) bottom from MMVU (Zhao et al. 2025) requires LVLMs to both recognize a chemical reaction in the video and perform numeric computations based on the question context. This *coupling* makes it difficult to pinpoint the error source—whether due to incorrect fact identification (*e.g.*, failing to detect the reaction) or faulty reasoning (*e.g.*, miscalculating). In contrast, `Video SimpleQA` **exclusively** focuses on fact identification, providing a clearer assessment of LVLMs' fact-grounding ability.[2]

- **Multi-hop fact decomposition**: As shown in Figure 2, `Video SimpleQA` includes not only the final multi-hop QA pairs but also the decomposed *per-fact sub-QAs*, enabling fine-grained evaluations. While some cases in MMVU (Zhao et al. 2025) also involve knowledge from multiple external sources, they do not provide such explicit per-fact decomposition, making it difficult to assess how each individual fact contributes to the final answer.

We conduct comprehensive evaluations of 33 state-of-the-art LVLMs on `Video SimpleQA`, revealing several critical insights: 1) **Significant performance gap**: Both proprietary and open-source LVLMs substantially underperform compared to human expertise; 2) **Overconfidence bias**: Most LVLMs exhibit systematic overconfidence in their predictions despite output inaccuracies, with notable variations in calibration quality (*c.f.* Figure 7); 3) **Efficiency-performance tradeoff**: Retrieval-Augmented Generation (RAG) yields significant gains at the cost of inference efficiency (*c.f.* Table 3); 4) **Multi-hop performance bottleneck**: Multi-hop QA performance significantly lags behind single-hop sub-tasks, with the initial video-grounded hop acting as the primary bottleneck (*c.f.* Table 4). More experimental findings are available in the supplementary material.

Related Work

Factuality Benchmarks. Factuality is the capability of LLMs to generate content that aligns with factual information, which can be substantiated by authoritative sources such as Wikipedia or textbooks (Akhtar et al. 2023; Wang et al. 2024c). Evaluating LLM factuality presents a non-trivial challenge and various benchmarks are proposed in the text-based (Lin, Hilton, and Evans 2022; Chern et al. 2023; Gou et al. 2023; Wei et al. 2024; He et al. 2024b) and image-based scenarios (Marino et al. 2019; Wang et al. 2017; Jain et al. 2021; Gu et al. 2025). As one of the pioneering works, TruthfulQA (Lin, Hilton, and Evans 2022) specifically targets imitative falsehoods in LLM-generated responses, which stem from erroneous preconceptions or knowledge gaps. Recently, the SimpleQA series of works (Wei et al. 2024; He et al. 2024b; Gu et al. 2025; Cheng et al. 2025) facilitate factuality evaluation by constraining the scope to short, fact-seeking questions with simple answers, making factuality assessment more tractable compared to previous long, open-ended model outputs. Despite of this, the community urgently needs a standard benchmark for trustworthy factuality evaluation *in video contexts*. To address this gap, our `Video SimpleQA` emerges.

Video Understanding Benchmarks. Recently, video benchmarks have been designed for evaluations in comprehensive tasks, including temporal perception (Li et al. 2024b), reasoning (Cai et al. 2024; Chen et al. 2024), navigation (Yang et al. 2024), long-form comprehension (Song et al. 2024; Chandrasegaran et al. 2024), *etc*. However, current video benchmarks largely overlook factuality evaluation, resulting in a lack of assessment for LVLMs' ability to generate factually accurate responses. Compared to video hallucination benchmarks (Wang et al. 2024d; Guan et al. 2024; Zhang et al. 2024a), which primarily assess models' adherence to video contents, our focused factuality evaluation emphasizes the model's alignment with verifiable external world knowledge (Wang et al. 2023, 2024c).

Differentiation from Knowledge-based and Discipline-base Benchmarks. Existing knowledge-based (Zhang et al. 2024b; Hu et al. 2025) and discipline-based benchmarks (Zhao et al. 2025; He et al. 2024a) either contain *hypothetical/subjective* reasoning (*e.g.*, the categories of `societal norms` and `social interactions` in WorldQA (Zhang et al. 2024b)) or *narrow their scopes* to single TV show (Garcia et al. 2020) or discipline-related knowledge (Zhao et al. 2025; He et al. 2024a). Our `Video SimpleQA` addresses these limitations by enforcing *objective* factuality verification and ensuring *diversity* across various categories. In addition, we introduce per-hop fact-grounded QA for *fine-grained* evaluations.

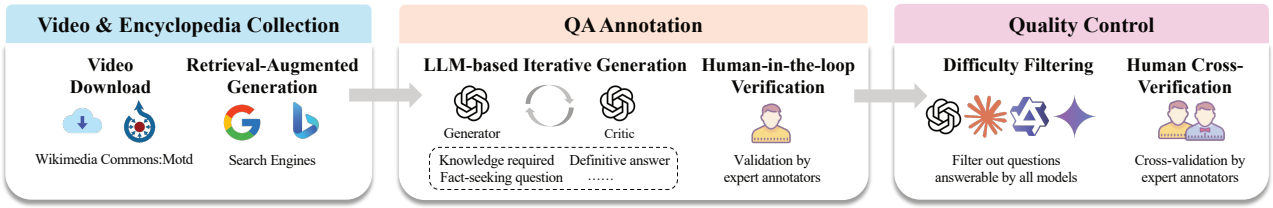


Figure 3: **An overview of the construction pipeline** of `Video SimpleQA`.

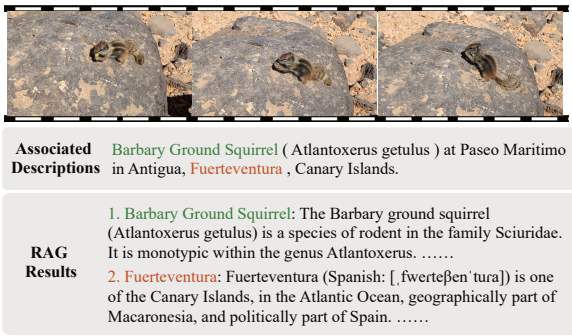


Figure 4: **The encyclopedia collection process** including the raw associated descriptions in Wikimedia and the RAG results[3] for the specialized terms extracted by GPT-4o.

Video SimpleQA

The construction pipeline of `Video SimpleQA` is illustrated in Figure 3, which includes video & encyclopedia collection, QA annotations, and quality control.

Video & Encyclopedia Collection

Video Collection: To ensure broad coverage, we curate the knowledge-intensive videos from the "Media of the Day" page of Wikimedia Commons[3] together with the accompanied brief descriptions or scientific illustrations. Note that files on the "Media of the Day" page are freely licensed, which avoids introducing any potential copyright concerns.

Encyclopedia Collection: As shown in Figure 4, although the associated textual descriptions on the Wikimedia page provide related descriptions, the explanations for the specialized terms (*e.g.*, `Barbary Ground Squirrel`, `Fuerteventura`) still lack formal definitions. To construct a more comprehensive encyclopedia, we leverage GPT-4o (OpenAI 2024) to extract key terms from the original descriptions and then obtain detailed explanations for these terms via RAG. Specifically, we apply LlamaIndex (Liu 2022) as the RAG method, with search results from Google and Bing as data sources.

QA Annotations

The annotation pipeline for `Video SimpleQA` follows a two-stage process: (1) automated LLM-based iterative generation and (2) human-in-the-loop verification refinement.

[3]https://commons.wikimedia.org/wiki/Commons:Media_of_the_day

LLM-based Iterative Generation: The iterative generation process involves two LLMs, a *generator* LLM for initial QA pair synthesis and a *critic* LLM for quality assessment. Specifically, the generator receives videos and encyclopedic knowledge to generate candidate QA pairs. Subsequently, the critic LLM systematically evaluates output compliance with predefined quality criteria, providing targeted feedback for refinement. This iterative process continues for up to three refinement cycles, with non-compliant outputs being discarded post-final iteration to ensure rigorous quality control. Both generator and critic are implemented as GPT-4o.

The explicit construction criteria are as follows: 1) *Knowledge required*: The questions should necessitate both video content and relevant external factual knowledge. Those that can be answered solely based on either source should be excluded. For example, two questions that should be eliminated are: `What color is the insect in the video?` (which relies solely on video content) and `Which president of the United States was Obama?` (which relies solely on external knowledge); 2) *Fact-seeking question*: The generated question should be factually grounded without any hypothetical or subjective reasoning; 3) *Definitive answer*: To ensure a rigorous evaluation, each question must have a single, unambiguous, and time-invariant answer. To achieve this, we explicitly define the level of granularity in question phrasing. For example, we use "which year" instead of "when" and "which city" instead of "where" to eliminate ambiguity; 4) *Short-form answer*: The answers should be in a short-form format; 5) *Multi-hop facts*: To answer the question, it requires involving multiple factual sources; 6) *Temporal grounded*: The questions are grounded in one or more video segments rather than specific frames.

Human-in-the-loop Verification: Through the iterative generation, we obtain QA annotations of reasonable quality. To further enhance the reliability, we train expert annotators to refine the LLM-generated QA annotations. The expert annotators are first required to watch the complete video and examine the collected encyclopedic knowledge. They then evaluate whether the LLM-generated QA annotations meet the specified criteria and manually revise them if necessary.

To ensure *multi-hop* fact grounding, annotators were additionally instructed to decompose each multi-hop QA into a series of sub-QAs (*c.f.* Figure 2). The decomposition follows two rules: 1) **Single fact per sub-QA**: Each sub-QA targets a single fact that is independently verifiable; 2) **Referential dependency**: Each sub-QA builds upon the answer to the previous one, forming the step-by-step fact chaining.

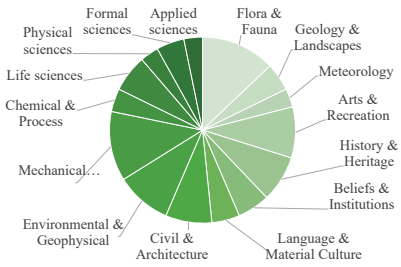
Table 2: **Evaluation results (%) of open-source and proprietary multi-modal LLMs on `Video SimpleQA`** . For metrics, CO, NA, IN, and CGA denote "Correct", "Not attempted", "Incorrect", and "Correct given attempted", respectively. For subtopics, ENG, NAT, SCI and SAC represent "Engineering", "Nature", "Science" and "Society and Culture".

Model	Overall results on 5 metrics					F-score on 4 primary categories			
	CO	IN↓	NA↓	CGA	F-score	ENG	NAT	SCI	SAC
Human Performance									
Human Open-book	87.0	5.0	8.0	94.6	90.6	85.2	83.7	89.1	96.8
Human Closed-book	59.0	18.5	22.5	76.1	66.5	58.4	52.8	54.2	80.6
Proprietary Multi-modal LLMs									
o4-mini (OpenAI 2025b)	53.7	45.3	0.9	54.2	54.0	44.3	59.4	56.8	54.0
o3 (OpenAI 2025b)	**66.3**	**33.6**	**0.1**	**66.4**	**66.3**	**63.0**	**71.3**	63.5	**68.8**
GPT-4.5 (OpenAI 2025a)	52.9	42.5	4.6	55.4	54.1	49.5	57.5	57.9	51.4
GPT-4o (OpenAI 2024)	47.7	45.9	6.4	51.0	49.3	45.1	57.1	52.7	45.4
Claude Sonnet 4 (Anthropic 2025b)	32.8	51.2	16.0	39.0	35.6	33.0	34.3	37.9	35.0
Claude 3.7 Sonnet (Anthropic 2025a)	32.6	47.3	20.1	40.8	36.2	24.2	40.3	41.9	34.5
Claude 3.5 Sonnet2 (Anthropic 2024)	33.7	58.2	8.1	36.7	35.2	26.1	37.2	38.5	35.4
Claude 3.5 Sonnet (Anthropic 2024)	31.5	53.5	15.0	37.0	34.0	26.8	36.0	38.1	32.5
Gemini 2.5 Pro (Comanici et al. 2025)	61.2	34.3	4.5	64.1	62.6	53.5	65.8	**67.1**	61.5
Gemini 2.5 Flash (Comanici et al. 2025)	53.7	34.9	11.3	60.6	57.0	46.0	61.6	61.4	56.2
Qwen-VL-Max (Bai et al. 2023)	39.2	57.1	3.7	40.7	39.9	27.4	46.2	48.8	35.1
Qwen-VL-Plus (Bai et al. 2023)	21.9	63.3	14.7	25.7	23.7	10.5	25.6	30.5	21.8
Open-source Multi-modal LLMs									
InternVL3-78B (Zhu et al. 2025)	33.7	65.6	**0.7**	33.9	33.8	25.4	41.2	38.6	30.6
InternVL3-38B (Zhu et al. 2025)	31.4	67.7	0.9	31.7	31.5	21.3	33.3	35.7	31.8
InternVL3-14B (Zhu et al. 2025)	24.9	73.3	1.8	25.4	25.2	14.6	32.3	28.4	24.6
InternVL3-9B (Zhu et al. 2025)	22.6	72.9	4.5	23.7	23.1	12.8	33.2	27.9	19.9
InternVL3-8B (Zhu et al. 2025)	23.3	75.2	1.5	23.7	23.5	16.2	30.7	25.6	22.4
Qwen2.5-VL-72B (Bai et al. 2025)	**38.7**	57.3	4.0	**40.3**	**39.5**	**26.1**	**47.0**	**48.2**	**34.7**
Qwen2.5-VL-32B (Bai et al. 2025)	30.3	67.1	2.7	31.1	30.7	18.1	39.3	37.4	27.0
Qwen2.5-VL-7B (Bai et al. 2025)	24.7	71.2	4.1	25.8	25.3	13.8	25.6	30.8	25.1
Qwen2-VL-72B (Wang et al. 2024a)	32.7	59.0	8.3	35.7	34.2	20.2	39.0	40.0	33.2
Qwen2-VL-7B (Wang et al. 2024a)	22.4	69.4	8.2	24.4	23.4	15.9	23.9	25.1	25.0
LLaVA-1.5-13B (Liu et al. 2023)	19.3	76.7	4.1	20.1	19.7	11.6	21.2	21.9	20.8
LLaVA-1.5-7B (Liu et al. 2023)	16.1	78.5	5.4	17.1	16.6	8.9	19.2	19.0	17.1
LLaVa-NeXT-Video-34B (Liu et al. 2024)	11.2	83.8	4.9	11.8	11.5	7.6	11.5	10.3	14.5
LLaVa-NeXT-Video-7B (Liu et al. 2024)	9.3	52.9	37.8	14.9	11.4	7.4	15.1	14.5	8.7
LLaVA-OneVision-72B (Li et al. 2024a)	25.4	73.6	1.0	25.7	25.5	15.9	25.3	28.5	27.3
LLaVA-OneVision-7B (Li et al. 2024a)	18.9	76.6	4.5	19.8	19.3	12.1	26.3	21.3	18.4
DeepSeek-VL2 (Wu et al. 2024)	3.2	49.1	47.7	6.1	4.2	3.0	4.4	3.0	5.9
DeepSeek-VL2-Small (Wu et al. 2024)	5.9	52.1	42.1	10.1	7.4	3.9	10.1	9.5	6.2
DeepSeek-VL2-Tiny (Wu et al. 2024)	16.1	75.6	8.3	17.6	16.8	9.6	27.1	17.5	16.0
Kimi-VL (Team et al. 2025a)	18.3	**44.4**	37.3	29.1	22.4	14.9	19.8	25.2	24.5
Keye-VL (Team et al. 2025b)	25.4	53.9	20.7	32.0	28.3	15.3	23.1	37.2	26.6

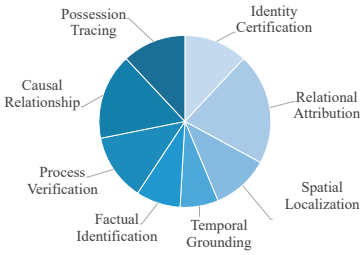
Quality Control

Difficulty Filtering. To ensure an appropriate level of assessment difficulty, we establish filtering rules to exclude questions that are easy to answer. In particular, questions correctly answered by all four state-of-the-art models, including GPT-4o (OpenAI 2024), Claude Sonnet 4 (Anthropic 2025b), Gemini 2.5 Pro (Comanici et al. 2025), and Qwen-VL-Max (Bai et al. 2023) are deemed insufficiently challenging and consequently excluded from our benchmark. This filtering strategy ensures our dataset maintains a sufficient level of difficulty for meaningful evaluations.

Human Cross-verification. To further enhance the dataset quality, a rigorous human validation process is implemented. Each question is independently evaluated by two annotators for compliance with our predefined criteria. Questions are discarded if either annotator deems them non-compliant. Meanwhile, annotators are required to verify answers against authoritative sources (such as Wikipedia). Finally, the final dataset undergoes security auditing to address potential security issues. All these stringent human verification processes ensure both the accuracy of our dataset and its adherence to established criteria.



(a)



(b)

Statistics of `Video SimpleQA`	Value
2/3/4-hop QA pairs	928 / 469 / 107
Total QA pairs	1504
Question Length (avg/max)	15.64 / 38
Answer Length (avg/max)	1.28 / 9
Unique Videos	1079
Video Length (Seconds, avg/max)	201 / 8763
Number of primary category	4
Number of secondary category	15
Number of tertiary category	84

(c)

Figure 5: (a) **Video distribution** at the secondary level; (b) **Question type distribution**; (c) **Key statistics**.

Dataset Statistics. The key statistics of `Video SimpleQA` are demonstrated in Table 5c. As shown, it consists of 1079 videos with 1504 expert-annotated QA pairs. The videos span 4 primary categories, 15 secondary categories and 84 tertiary categories. The average lengths of questions and answers are 15.64 and 1.28 words, respectively, aligning with our intended short-form design. The video distribution at the secondary level is demonstrated in Figure 5a. The question type distribution is visualized in Figure 5b.

Experiments

Experimental Setup

Evaluated Models. We benchmark comprehensive state-of-the-art LLMs, including **12 proprietary models**, including o4-mini, o3, GPT-4.5, GPT-4o, Claude Sonnet 4, Claude 3.7 Sonnet, Claude 3.5 Sonnet series, Gemini 2.5 series, and Qwen-VL series, and **21 open-source models**, including InternVL3 series, Qwen2.5-VL series, Qwen2-VL series, LLaVA-1.5 series, LLaVA-NeXT-Video series, LLaVA-OneVision series, DeepSeek-VL2 series, Kimi-VL, and Keye-VL. Following Video-MME (Fu et al. 2024), we maximize frame utilization of each model by inputting the maximum frames that fit within its context window.

Evaluation Metrics. Following SimpleQA (Wei et al. 2024), we set up five evaluation metrics: (1) **Correct**: The predicted answer comprehensively contains all key information from the reference answer while containing no contradictory elements. (2) **Incorrect**: The predicted answer contradicts the reference answer. The indirect or equivocal responses (*e.g.*, "possibly", "I think, although I'm not sure") are also considered incorrect. (3) **Not attempted**: The reference answer is not fully given in the predicted answer, and no statements in the answer contradict the gold target. (4) **Correct given attempted**: The ratio of correctly answered questions among attempted ones. (5) **F-score**: The harmonic mean values between *correct* and *correct given attempted* metrics. We follow the paradigm of *LLM-as-a-Judge* (Gu et al. 2024) and employ `o3` as the judge model.

Experimental Findings[4]

The evaluation results on `Video SimpleQA` are presented in Table 2, and key findings are summarized as follows:

[4]More experiments are available in the supplementary material.

`Video SimpleQA` is challenging: To assess human performance on `Video SimpleQA` , we sample 200 instances and recruit five participants to independently complete the tasks under two distinct conditions: with access to external resources (*e.g.*, Internet, textbooks) and without such access. These configurations correspond to the *human open-book* and *human closed-book* settings documented in Table 2.

Compared to the human open-book performance, both open-source and proprietary models demonstrate suboptimal performance. Specifically, the top-performing proprietary model, o3 (OpenAI 2025b), achieves an F-score of 66.3%. Open-source models exhibit even poorer results, with the best-performing one, Qwen2.5-VL-72B (Wang et al. 2024a) attaining only 39.5% F-score. This demonstrates that LVLMs still exhibit limited capability in factuality adherence within video contexts, while also highlighting the necessity of establishing `Video SimpleQA` .

LVLMs are overconfident in what they generate: All models exhibit higher IN values (incorrect predictions) than NA values (non-attempted responses), indicating a prevalent tendency to generate answers despite insufficient factual knowledge. To further investigate this overconfidence phenomenon, we conduct *calibration* experiments (Guo et al. 2017) to examine whether language models "know what they know", *i.e.*, whether the model's assessed confidence scores align with the actual likelihood of its responses being correct. Specifically, we instruct LVLMs to self-assess confidence scores (0-100) for their predictions. Responses are grouped into confidence intervals (10-point bins), and we calculate *interval accuracy* (correct predictions per bin). As shown in Figure 7, except for o3, which demonstrates superior calibration, all other models *mostly* fall below the perfect calibration line, indicating systematic overconfidence.

RAG yields significant gains at the cost of inference efficiency: We explore RAG to facilitate `Video SimpleQA` benchmark comprehension in a three-step approach: 1) Prompting GPT-4o with video and questions to extract key textual entities; 2) Applying LlamaIndex with Google and Wikipedia as sources to retrieve relevant information based on these extracted key entities; 3) Augmenting the input question with the retrieved information.

As shown in Table 3, RAG achieves consistent and significant F-score improvements over vanilla models. For instance, when integrated with Claude Sonnet 4 (Anthropic

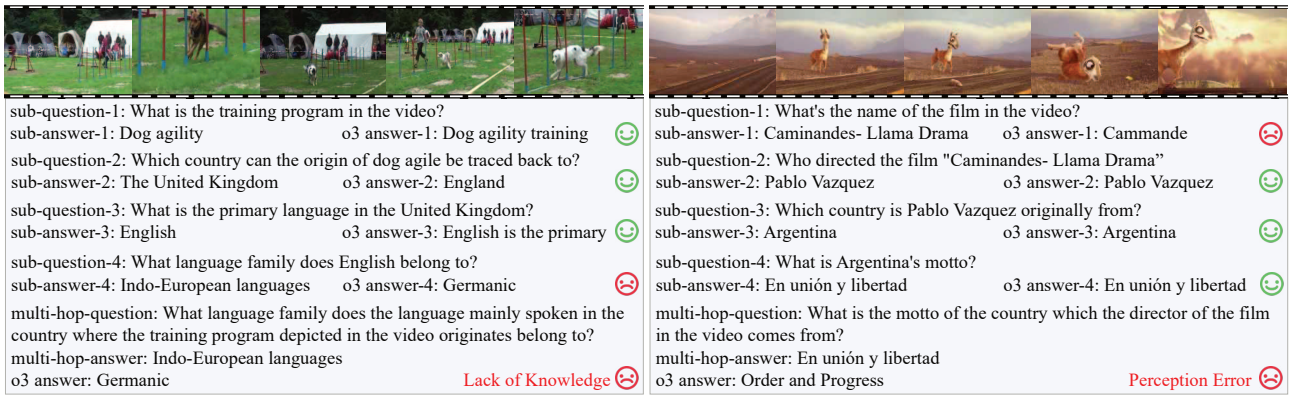


Figure 6: **Visualizations** of per-hop evaluation results of o3 (OpenAI 2025b).

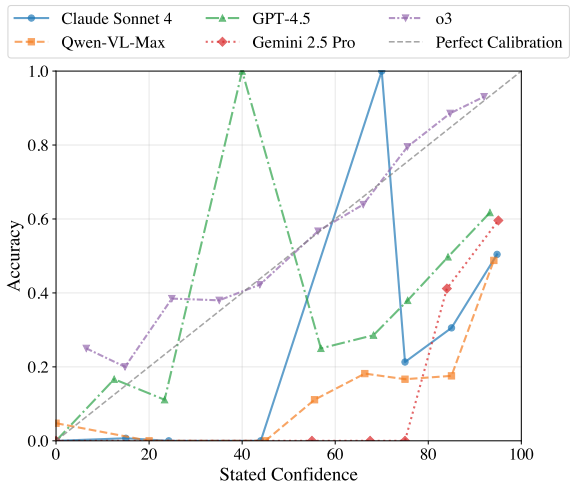


Figure 7: **Calibration curves** based on the self-stated confidence scores and interval-level accuracy.

2025b), RAG delivers an absolute improvement of 23.2% (35.6% *vs*. 58.8%). However, this performance gain comes with substantial computational overhead. Table 3 also quantifies the total inference time, demonstrating that RAG significantly impairs inference efficiency. Our findings highlight the critical trade-off between performance gains and computational practicality.

Per-hop factual evaluation: In addition to multi-hop factual QA, our `Video SimpleQA` benchmark also incorporates decomposed per-fact sub-QAs to facilitate fine-grained evaluations. As shown in Table 4, we present F-scores for both the final multi-hop QA and the sub-QAs across all 4-hop questions in `Video SimpleQA` . Our analysis reveals: 1) **Multi-hop challenge**: The final multi-hop QAs achieve substantially lower F-scores than most single-hop sub-QAs, underscoring the difficulty of multi-hop fact grounding; 2) **First-hop bottleneck**: The F-score for the first hop is markedly lower than those of later hops, likely due to its reliance on accurate object or event recognition, which poses a key challenge. In contrast, LVLMs perform better on subsequent hops (Q2–Q4) given clearer contextual grounding.

Table 3: **Comparisons between vanilla models and models with RAG** by F-score (%) and the inference time (min).

Model	F-score		Inference Time	
	vanilla	*w/* RAG	vanilla	*w/* RAG
o3	66.3	**69.0**	**27.8**	54.7
GPT-4o	49.3	**61.3**	**30.1**	53.9
Gemini 2.5 Pro	62.6	**66.2**	**33.2**	60.1
Claude Sonnet 4	35.6	**58.8**	**29.9**	56.4
Qwen-VL-Max	39.9	**57.0**	**24.2**	61.2

Table 4: **Per-hop factual evaluations** for 4-hop questions in terms of F-score (%). Q1-Q4 denote the decomposed per-hop questions. Refer to Figure 6 for an illustrative case.

Model	QA1	QA2	QA3	QA4	Multi-hop
o3	**74.9**	**89.7**	**95.3**	**88.8**	**78.5**
GPT-4o	59.1	85.4	92.5	80.9	47.0
Claude Sonnet 4	43.5	85.7	80.8	61.6	47.7
Gemini 2.5 Pro	65.1	78.7	60.4	28.3	69.8
Qwen-VL-Max	37.9	69.2	83.6	72.3	40.7

In Figure 6, we visualize the per-hop evaluation results of the o3 model. This allows us to clearly identify which specific piece of factual knowledge the model lacks. For instance, in the left case, o3 lacks knowledge about the language family of English, while in the right case, it fails to recognize the film Caminandes: Llama Drama.

Conclusions

We present `Video SimpleQA`, the first benchmark explicitly designed for evaluating factual grounding in video contexts. Distinct from prior works, our framework introduces the following diagnostic dimensions: knowledge integration, multi-hop fact-seeking questioning, short-form definitive answering, and temporal grounded demands. Through the extensive evaluation of 33 state-of-the-art LVLMs, we reveal notable deficiencies in factual adherence, uncover prevalent model overconfidence, trade-offs associated with RAG, and the critical performance bottleneck.

Supplementary Material

This **supplementary material** is organized as follows. We begin with a detailed description of the experimental setups, then present additional experimental findings, followed by the analysis of error types and the visualization results.

Specifically, the **experiment setups** include the following aspects:

- Experimental configurations.
- Evaluation prompts.
- Video distributions.
- Biographies of annotators.
- Copyright clarifications.

More experimental findings include the following aspects:

- Results with different judge models.
- Performance across secondary categories.
- Model size scaling remains effective.
- Frame number scaling remains effective.
- Performance of smaller LVLMs.
- Test-time compute yields limited benefits.
- Temporal scope analysis.
- More per-hop factual evaluation results.

Experiment Setup

Experimental configurations. Table 5 details the configuration of each evaluated model. We used the default settings from the official implementation of each model. All experiments were reproducible on a workstation equipped with 8 NVIDIA A100 GPUs. Across all experiments, the temperature is set to 1.0, with a maximum output length of 1024 tokens. Each experiment was repeated three times to ensure reproducibility and statistical reliability.

Evaluation prompts. The prompts for the grader, along with instructions guiding the model to output answers and confidence scores, are illustrated in Figure 17, Figure 18 and Figure 19, respectively.

Video distributions. In Table 10 and Table 11, we present the video taxonomy distributions of `Video SimpleQA` including 4 primary categories, 15 secondary categories, and 84 tertiary categories.

Biographies of annotators. The detailed information of annotators who participated in the construction of `Video SimpleQA` can be found in Table 9. All annotators come from universities ranked in the top 500 of the 2026 QS Global Rankings[5], and they are all fluent in English.

Copyright clarifications. The videos of our `Video SimpleQA` benchmark are sourced from the "Media of the Day" page of Wikipedia, which are freely licensed. Therefore, the construction of `Video SimpleQA` *avoids introducing any potential copyright concerns*.

To better align with ethical standards, we have implemented the following measures: 1) **Data desensitization**: To reduce possible security threats during the evaluation stage, the finalized dataset was independently reviewed by six professional security auditors. Each data instance was examined by at least two reviewers, focusing on potential issues such as gender bias, offensive language, or politically sensitive content. Only the samples that passed all checks were included in the final release; 2) **Non-commercial research**: We release our code and dataset under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) License. They are available strictly for non-commercial research; 3) **Opt-out mechanism**: We release our dataset with an opt-out mechanism, allowing content owners and individuals appearing in the videos to request the removal of their video references.

[5]https://www.topuniversities.com/world-university-rankings

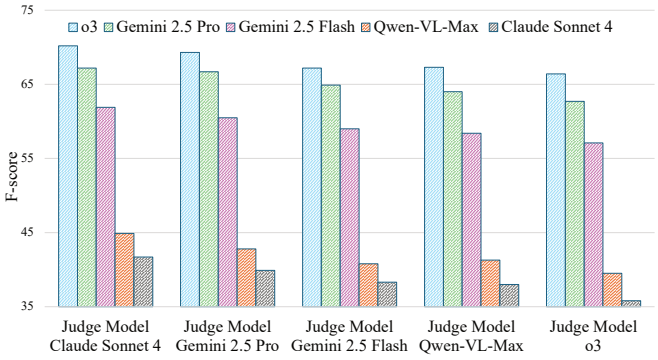


Figure 8: Results with different judge models.

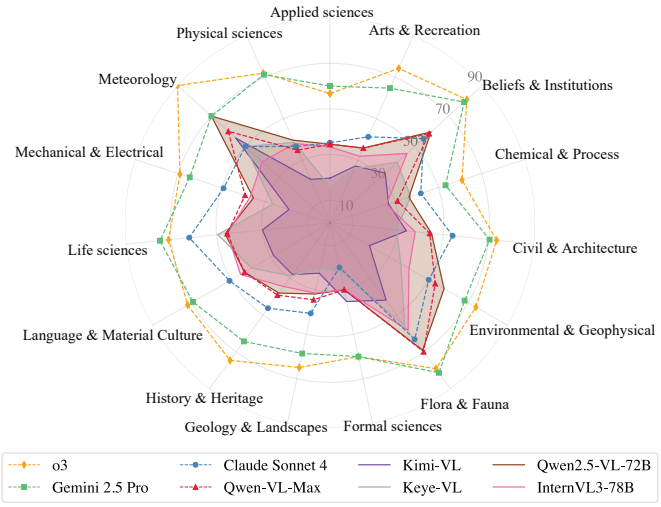


Figure 9: The performance of different models across 15 secondary categories in `Video SimpleQA` .

More Experimental Findings

Results with different judge models. The short-form answer paradigm of `Video SimpleQA` enables automated evaluation through LLM-as-a-judge frameworks with low run-to-run variance. To demonstrate this, we select five typical LVLMs, and evaluate them using various judge models including o3 (OpenAI 2025b), Gemini 2.5 Pro (Comanici

Table 5: **Configurations** of evaluated LVLMs in Video SimpleQA.

Organization	Model	Release	Version	Input Frames
		Proprietary Multi-modal LLMs		
OpenAI	o4-mini	2025-4	o4-mini-2025-04-16	32
	o3	2025-4	o3-2025-04-16	32
	GPT-4.5	2025-2	gpt-4.5-2025-02-27	32
	GPT-4o	2024-5	gpt-4o-2024-08-06	32
Anthropic	Claude Sonnet 4	2025-5	claude-sonnet-4	32
	Claude 3.7 Sonnet	2025-2	claude-3.7-sonnet	16
	Claude 3.5 Sonnet2	2024-10	claude-3.5-sonnet-v2	16
	Claude 3.5 Sonnet	2024-6	claude-3.5-sonnet	16
Google	Gemini 2.5 Pro	2025-3	gemini-2.5-pro-2025-06-17	32
	Gemini 2.5 Flash	2025-3	gemini-2.5-flash-2025-06-17	32
Alibaba	Qwen-VL-Max	2024-1	qwen-vl-max	32
	Qwen-VL-Plus	2023-11	qwen-vl-plus	32
		Open-source Multi-modal LLMs		
Shanghai AI Lab	InternVL3-78B	2025-4	InternVL3-78B	16
	InternVL3-38B	2025-4	InternVL3-38B	16
	InternVL3-14B	2025-4	InternVL3-14B	16
	InternVL3-9B	2025-4	InternVL3-9B	16
	InternVL3-8B	2025-4	InternVL3-8B	16
	InternVL3-2B	2025-4	InternVL3-2B	16
	InternVL3-1B	2025-4	InternVL3-1B	16
Alibaba	Qwen2.5-VL-72B	2025-2	Qwen2.5-VL-72B-Instruct	32
	Qwen2.5-VL-32B	2025-2	Qwen2.5-VL-32B-Instruct	32
	Qwen2.5-VL-7B	2025-2	Qwen2.5-VL-7B-Instruct	32
	Qwen2.5-VL-3B	2025-2	Qwen2.5-VL-3B-Instruct	32
	Qwen2-VL-72B	2024-8	Qwen2-VL-72B-Instruct	16
	Qwen2-VL-7B	2024-8	Qwen2-VL-7B-Instruct	16
	Qwen2-VL-2B	2024-8	Qwen2-VL-2B-Instruct	16
Llava Hugging Face	LLaVA-1.5-13B	2023-9	llava-1.5-13b-hf	4
	LLaVA-1.5-7B	2023-9	llava-1.5-7b-hf	4
	LLaVA-NeXT-Video-34B	2024-4	LLaVA-NeXT-Video-34B-hf	8
	LLaVA-NeXT-Video-7B	2024-4	LLaVA-NeXT-Video-7B-hf	4
	LLaVA-OneVision-72B	2024-8	llava-onevision-qwen2-72b-ov-hf	16
	LLaVA-OneVision-7B	2024-8	llava-onevision-qwen2-7b-ov-hf	16
	LLaVA-OneVision-0.5B	2024-8	llava-onevision-qwen2-0.5b-ov-hf	16
DeepSeek	DeepSeek-VL2	2024-12	deepseek-vl2	2
	DeepSeek-VL2-Small	2024-12	deepseek-vl2-small	2
	DeepSeek-VL2-Tiny	2024-12	deepseek-vl2-tiny	2
MoonshotAI	Kimi-VL	2025-4	Kimi-VL-A3B-Instruct	32
Kwai-Keye	Keye-VL	2025-6	Keye-VL-8B-Preview	32

et al. 2025), Gemini 2.5 Flash (Comanici et al. 2025), Qwen-VL-Max (Bai et al. 2023), and Claude Sonnet 4 (Anthropic 2025b). As shown in Figure 8, while the specific scores from different judge models vary, the relative rankings of the evaluated models remain consistent. This demonstrates the robustness of our evaluation under the short-form paradigm.

Performance across secondary categories. Figure 9 demonstrates the F-score performance across 15 secondary categories. As shown, we observe distinct performance patterns among the compared LVLMs.

- *Capability distribution*: o3 (OpenAI 2025b) demonstrates the most consistent performance with superior F-scores across domains, particularly excelling in the category of Meteorology. Gemini 2.5 Pro (Comanici et al. 2025) follows with complementary strengths and exhibits notable advantages in the category of Life sciences compared to other LVLMs.
- *Imbalanced performance profiles*: Qwen2.5-VL-72B (Bai et al. 2025) and Qwen-VL-MAX (Bai et al. 2023) show significant performance variance, with Qwen-VL-MAX (Bai et al. 2023) severely underperforming in Formal Sciences compared to its moderate capabilities in other domains. Kimi-VL (Team et al. 2025a) and Keye-VL (Team et al. 2025b) consistently underper-

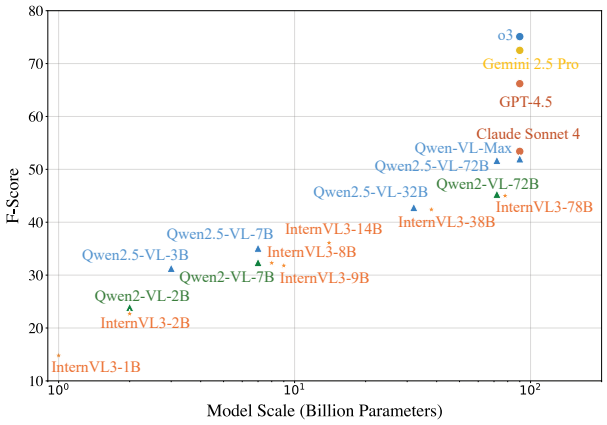


Figure 10: The relationship between model size and F-score.

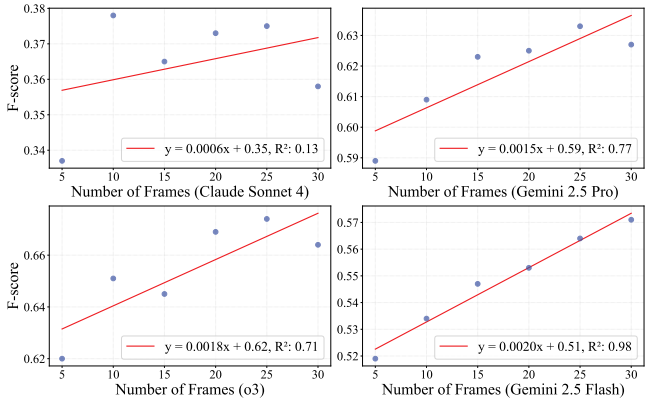


Figure 11: Frame number scaling experiments.

form across most domains, with particularly low F-scores in the technical areas (*e.g.*, `Physical Sciences`, `Applied Sciences`).

- *Disciplinary performance gap*: All models exhibit systematically lower F-scores in scientific domains (*e.g.*, `Formal/Applied Sciences`) compared to humanities-oriented categories (History & Heritage, Civil & Architecture), with Kimi-VL (Team et al. 2025a) showing the most pronounced disparities.

Model size scaling remains effective: As evidenced by the F-score across various model sizes in Figure 10, model size scaling continues to demonstrate effectiveness, where larger architectures exhibit consistently superior performance. This observation aligns with the widely recognized scaling law principle.

Frame number scaling remains effective: In Figure 11, we demonstrate the impact of the number of sampled frames on performance. The results for Gemini 2.5 Pro, o3, and Gemini 2.5 Flash reveal a positive correlation between the video frame number and the F-score, thereby validating the effectiveness of frame number scaling. For the results of Claude Sonnet 4, the optimum result is achieved when setting the frame number to 10.

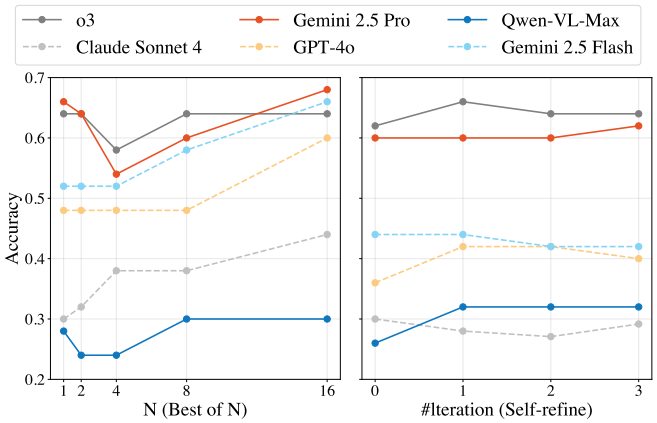


Figure 12: **Evaluations of test-time compute** including Best-of-N and Self-refine.

Performance of smaller LVLMs: In addition to the evaluation results of the 33 LVLMs presented in the main paper, we have also included results for smaller models (with parameter counts of 0.5B, 1B, 2B and 3B) in Table 6. This serves as a valuable reference for users with limited resources who wish to utilize the proposed `Video SimpleQA` dataset.

Test-Time compute yields limited benefits: We empirically investigate the effectiveness of test-time compute (Snell et al. 2024) strategies on `Video SimpleQA` by conducting experiments with 200 randomly sampled instances. Two approaches are evaluated: 1) **Best-of-N**: The model independently generates N responses and selects the one it considers the best as the final answer; 2) **Self-refine**: The model is prompted to iteratively refine the initial outputs using self-generated feedback (Madaan et al. 2023).

Figure 12 presents the accuracy (*i.e.*, the proportion of correct answers) under varying inference-time N (for Best-of-N) and different refinement iterations (for Self-refine). Experimental results reveal that both strategies fail to produce significant or consistent accuracy improvements. In some cases, these strategies even degrade performance. For instance, when increasing N from 2 to 4 in Best-of-N trials, o3 (OpenAI 2025b) and Gemini-2.5-Pro (Reid et al. 2024) exhibit reduced accuracy, suggesting that these models struggle to reliably select the best answer from multiple inferences. These findings highlight the challenges in improving factuality through post-hoc test-time compute strategies.

Temporal scope analysis: `Video SimpleQA` is designed to be temporally grounded, *i.e.*, answering questions should refer to one or more temporal segments in the video, rather than relying on a single frame. To demonstrate this, we randomly sampled 200 QA pairs and instructed expert annotators to categorize them based on the *necessary temporal scope* needed for accurate answers. We categorize the results as follows: 1) *short-term* scope required (<10 seconds); 2) *medium-term* scope required (10s-1min); or 3) *long-term* scope required (>1min). It should be emphasized that our definitions of short/medium/long-term specifically denote

Table 6: **Evaluation results (%) of small LVLMs on `Video SimpleQA`** . For metrics, CO, NA, IN, and CGA denote "Correct", "Not attempted", "Incorrect", and "Correct given attempted", respectively. For subtopics, ENG, NAT, SCI and SAC represent "Engineering", "Nature", "Science" and "Society and Culture".

Model	Overall results on 5 metrics					F-score on 4 primary categories			
	CO	IN↓	NA↓	CGA	F-score	ENG	NAT	SCI	SAC
InternVL3-2B (Zhu et al. 2025)	15.5	80.5	4.0	16.1	15.8	10.9	23.4	16.4	14.8
InternVL3-1B (Zhu et al. 2025)	11.3	82.5	6.1	12.1	11.7	6.1	19.2	12.1	11.2
Qwen2.5-VL-3B (Bai et al. 2025)	22.3	74.5	3.2	23.0	22.6	12.1	30.3	29.2	18.6
Qwen2-VL-2B (Wang et al. 2024a)	16.3	73.3	10.4	18.2	17.2	12.9	24.9	16.4	17.4
LLaVA-OneVision-0.5B (Li et al. 2024a)	7.8	85.7	6.5	8.3	8.0	6.1	11.6	5.8	10.0

Table 7: **The proportion and performance** (%) of QA pairs by different temporal scopes.

Type	Prop	CO	IN↓	NA↓	CGA	F-score
Short-term	**50.5**	**56.6**	**33.1**	**10.3**	**63.1**	**59.7**
Medium-term	32.1	48.5	38.8	12.7	55.6	51.8
Long-term	17.4	43.2	41.5	15.3	51.0	46.8

the temporal scope required to correctly answer QA pairs, distinct from the absolute video duration referenced in existing long-form video understanding benchmarks (Wang et al. 2024b; Fu et al. 2024).

Table 7 summarizes the proportion and performance under different temporal scopes, which reveals that 82.6% of cases require either short-term or medium-term temporal understanding to answer correctly. This distribution demonstrates that our `Video SimpleQA` benchmark indeed necessitates temporal understanding capabilities rather than simple frame-level analysis. Furthermore, as indicated in Table 7, videos with long-term temporal scope exhibit significantly lower performance metrics compared to the other two categories, which highlights the importance of long-context temporal modeling.

More per-hop factual evaluation results: In the main paper, we have presented the performance of per-fact sub-QA for all the 4-hop questions `Video SimpleQA` . Here, we present the per-fact sub-QA performance across all 3-hop questions. As shown in Table 8, we still observe that final multi-hop QAs achieve substantially lower F-scores than most single-hop sub-QAs, and the performance of the first hop is markedly lower than that of subsequent hops.

We observe that the accuracy of multi-hop questions is higher than that of sub-QA1. Through analysis, we found this occurs because in certain cases, even if sub-QA1 is answered incorrectly, both the ground-truth answer and the incorrect answer lead to a consistent final answer for the multi-hop question. For example in Figure 14, for the model o3, the `kata` type is incorrectly identified as `Heian Nidan`. Yet, whether based on this incorrect answer or the correct answer (`Heian Godan`), the final answer for the multi-hop question is consistently `Japan`. This further highlights the value of our proposed fine-grained per-hop evaluation, *i.e.*, it requires the model not only to correctly answer the final multi-hop question, but also to accurately answer each single-hop, fact-based question.

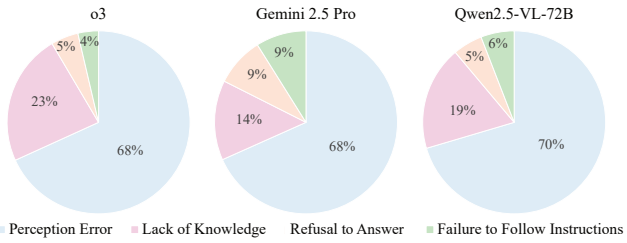


Figure 13: **Error type distributions** across o3 (OpenAI 2025b), Gemini 2.5 Pro (Comanici et al. 2025) and Qwen2.5-VL-72B (Bai et al. 2025).

Error Analysis

This section presents a case study analyzing error patterns in o3 (OpenAI 2025b), Gemini 2.5 Pro (Comanici et al. 2025) and Qwen2.5-VL-72B (Bai et al. 2025) through a systematic examination of all available samples per model spanning diverse question types. We categorize observed errors into four primary classes, each illustrated with representative examples:

- **Perception error:** Incorrect identification of objects. This occurs when LVLMs misidentify or fail to detect key visual elements in input videos (*c.f*. Figure 15 (a)).
- **Lack of knowledge:** Correct identification but lacking relevant knowledge. LVLMs accurately perceive the visual content but cannot provide accurate information due to knowledge limitations (*c.f*. Figure 15 (b)).
- **Refusal to answer:** LVLMs recognize their inability to make a confident determination and opt to abstain from answering (*c.f*. Figure 16 (c)).
- **Failure to follow instructions:** LVLMs understand the input but fail to properly execute the given instructions. This typically manifests in two ways: 1) the generated outputs do not conform to the specified format requirements, or 2) the responses are irrelevant to the question posed, *e.g*., addressing different topics than what was actually requested (*c.f*. Figure 16 (d)).

Table 8: **Per-hop factual evaluations** for 3-hop questions in terms of F-score (%). Q1-Q3 denote the decomposed per-hop questions.

Model	QA1	QA2	QA3	Multi-hop
o3	**64.9**	**94.5**	**90.5**	**74.9**
GPT-4o	55.0	88.0	87.4	56.7
Claude Sonnet 4	39.7	83.3	76.1	54.6
Gemini 2.5 Pro	63.4	81.7	59.5	71.7
Qwen-VL-Max	42.6	77.5	77.5	49.5

Visualizations

Evaluation results: The visualization results of three typical LVLMs (o3 (OpenAI 2025b), Gemini 2.5 Pro (Comanici et al. 2025) and Qwen2.5-VL-72B (Bai et al. 2025)) are illustrated in Figure 22 – Figure 26.

Limitations of existing discipline-based methods: As analyzed in the main paper, current discipline-based methods suffer from the following issues: 1) *Inclusion of hypothetical or subjective speculation*: For instance, in Figure 20, the questions often contain hypothetical or subjective expressions such as "might", "infer", "seem to", or "probably". In contrast, our `Video SimpleQA` requires answers to be crafted in an unambiguous and definitively correct manner; 2) *Entangling fact grounding and reasoning*: To answer the questions in the examples shown in Figure 21, the model needs to possess reasoning abilities such as matrix computation, minimum spanning tree planning, and time complexity analysis. This makes it difficult to independently evaluate the model's fact-grounding capability. In contrast, our `Video SimpleQA` exclusively focuses on fact identification, providing a clearer assessment of LVLMs' fact-grounding ability.

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sub-question-1: Which kata is the person performing in the video?
sub-answer-1: Heian Godan
o3-answer-1: Heian Nidan ☹

sub-question-2: Which Okinawan karate master developed the series of kata that includes Heian Godan?
sub-answer-2: Anko Itosu
o3-answer-2: Anko Itosu ☺

sub-question-3: Which country is Anko Itosu from?
sub-answer-3: Japan
o3-answer-3: Japan ☺

multi-hop-question: Which country did the Okinawan karate master who developed the series of kata being performed in the video come from?
multi-hop-answer: Japan
o3-answer: Japan ☺



sub-question-1: What insects are in the video?
sub-answer-1: Box Bug
o3-answer-1: Leaf-footed bugs (family Coreidae) ☹

sub-question-2: What order does the Box bug belong to?
sub-answer-2: Hemiptera
o3-answer-2: Hemiptera ☺

sub-question-3: What phylum does Hemiptera belong to?
sub-answer-3: Arthropoda
o3-answer-3: Arthropoda ☺

multi-hop-question: What phylum does that The order of insects in the video belong to?
multi-hop-answer: Arthropoda
o3-answer: Arthropoda ☺

Figure 14: Case studies where the incorrect sub-QA1 still yields the correct final multi-hop answer.



sub-question-1: What is the scientific name of the animal shown in the video?
sub-answer-1: Bufo bufo o3 answer-1: Rana temporaria

sub-question-2: To which taxonomic family does the European common toad, Bufo bufo, belong?
sub-answer-2: Bufonidae o3 answer-2: Bufonidae

multi-hop-question: To which taxonomic family does the animal shown in the video belong?
multi-hop-answer: Bufonidae o3 answer: Ranidae

error type: Perception Error
error reason: The model misidentified the animal in the video as a frog instead of a toad, leading to incorrect taxonomic classification.

(a)



sub-question-1: Which movie is this scene from in the video?
sub-answer-1: What Happened on Twenty-Third Street, New York City
o3 answer-1: What Happened on Twenty-Third Street, New York City

sub-question-2: What is the release month of the movie "What Happened on Twenty-Third Street, New York City"?
sub-answer-2: August 1901 o3 answer-2: July

multi-hop-question: What is the release month of the movie clip shown in the video?
multi-hop-answer: August 1901 o3 answer: July

error type: Lack of Knowledge
error reason: The model had incorrect factual information about the movie's release date, believing it was released in July instead of August 1901.

(b)

Figure 15: **Visualizations of typical error types** including (a) perception error; (b) lack of knowledge; (c) refusal to answer; (d) failure to follow instructions: part 1.



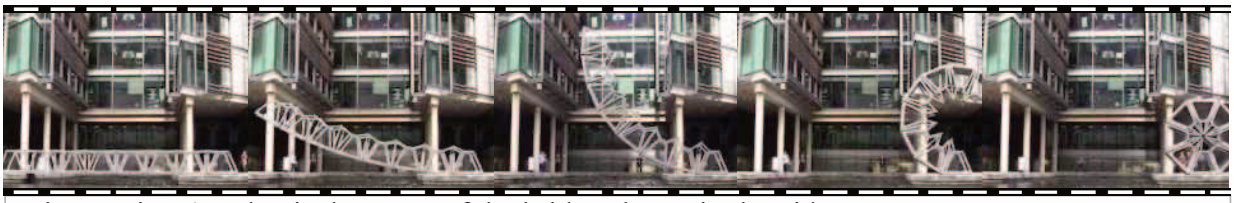
sub-question-1: What is the grain in the video?
sub-answer-1: Rice
Gemini 2.5 Pro answer-1: The grain being threshed in the video is most likely rice.

sub-question-2: What is the scientific name of rice?
sub-answer-2: Oryza sativa Gemini 2.5 Pro answer-2: Oryza sativa

multi-hop-question: What is the scientific name of the grain in the video?
multi-hop-answer: Oryza sativa
Gemini 2.5 Pro answer: The specific type of grain cannot be identified.

error type: Refusal to Answer
error reason: The model was overly cautious and refused to identify the grain type despite having correctly identified it as rice in the sub-questions.

(c)



sub-question-1: What is the name of the bridge shown in the video?
sub-answer-1: The Rolling Bridge Qwen2.5-VL-72B answer-1: Rolling Bridge

sub-question-2: What is the total length of The Rolling Bridge?
sub-answer-2: 12 metres Qwen2.5-VL-72B answer-2: 12 metres

multi-hop-question: What is the total length of the bridge shown in the video?
multi-hop-answer: 12 metres
Qwen2.5-VL-72B answer: The bridge in the video is Rolling Bridge.

error type: Failure to Follow Instructions
error reason: The model answered with the bridge name instead of the requested length measurement, failing to properly integrate information from sub-questions.

(d)

Figure 16: **Visualizations of typical error types** including (a) perception error; (b) lack of knowledge; (c) refusal to answer; (d) failure to follow instructions: part 2.

Your job is to look at some video frames generated from the video, a question generated from the video, a gold target, and a predicted answer, and then assign a grade of either ["CORRECT", "INCORRECT", "NOT_ATTEMPTED"].
First, I will give examples of each grade, and then you will grade a new example.

The following are examples of CORRECT predicted answers.
```
Question: What is the name of the man's child in the video?
Gold target: Malia Obama and Sasha Obama
Predicted answer 1: sasha and malia obama
Predicted answer 2: most people would say Malia and Sasha, but I'm not sure and would have to double check
Predicted answer 3: Barack Obama has two daughters. Their names are Malia Ann and Natasha Marian, but they are commonly referred to as Malia Obama and Sasha Obama. Malia was born on July 4, 1998, and Sasha was born on June 10, 2001.
```
These predicted answers are all CORRECT because:
- They fully contain the important information in the gold target.
- They do not contain any information that contradicts the gold target.
- Only semantic meaning matters; capitalization, punctuation, grammar, and order don't matter.
- Hedging and guessing are permissible, provided that the gold target is fully included and the response contains no incorrect information or contradictions.

The following are examples of INCORRECT predicted answers.
```
Question: What is the name of the man's child in the video?
Gold target: Malia and Sasha
Predicted answer 1: Malia.
Predicted answer 2: Malia, Sasha, and Susan.
Predicted answer 3: Barack Obama does not have any children.
Predicted answer 4: I think it's either Malia and Sasha. Or it could be Malia and Jackie. Or it could be Joey and Malia.
Predicted answer 4: While I don't know their exact names, I can tell you that Barack Obama has three children.
Predicted answer 5: It's possible you may mean Betsy and Olivia. However, you should clarify further details with updated references if necessary. Is that the correct answer?
Predicted answer 6: It may be the case that Obama's child is named James. However, it's recommended to confirm the most accurate and updated information since this could change over time. This model may not always reflect the most current information.
```
These predicted answers are all INCORRECT because:
- A factual statement in the answer contradicts the gold target. Incorrect statements that have some hedging (e.g., "it is possible that", "although i'm not sure, i think") are also considered incorrect.

The following are examples of NOT_ATTEMPTED predicted answers.
```
Question: What is the name of the man's child in the video?
Gold target: Malia and Sasha
Predicted answer 1: I don't know.
Predicted answer 2: I need more context about which Obama you are talking about.
Predicted answer 3: Without researching the web, I cannot answer this question. However, I can tell you that Barack Obama has two children.
Predicted answer 4: Barack Obama has two children. I know that one of them is Malia, but I'm not sure about the other one.
```
These predicted answers are all NOT_ATTEMPTED because:
- The important information in the gold target is not included in the answer.
- No statements in the answer contradict the gold target.

Figure 17: **Prompt** for grading: Part 1.

Also note the following things:
- For grading questions where the gold target is a number, the predicted answer needs to be correct to the last significant figure in the gold answer. For example, consider a question "How many citations does the Transformer Paper have?" with gold target "120k".
- Predicted answers "120k", "124k", and 115k" are all CORRECT.
- Predicted answers "100k" and "113k" are INCORRECT.
- Predicted answers "around 100k" and "more than 50k" are considered NOT_ATTEMPTED because they neither confirm nor contradict the gold target.
- The gold target may contain more information than the question. In such cases, the predicted answer only needs to contain the information that is in the question.
- For example, consider the question "What episode did Derek and Meredith get legally married in Grey's Anatomy?" with gold target "Season 7, Episode 20: White Wedding". Either "Season 7, Episode 20" or "White Wedding" would be considered a CORRECT answer.
- Do not punish predicted answers if they omit information that would be clearly inferred from the question.
- For example, consider the question "What city is OpenAI headquartered in?" and the gold target "San Francisco, California". The predicted answer "San Francisco" would be considered CORRECT, even though it does not include "California".
- Consider the question "What award did A pretrainer's guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity win at NAACL '24?", the gold target is "Outstanding Paper Award". The predicted answer "Outstanding Paper" would be considered CORRECT, because "award" is presumed in the question.
- For the question "What is the height of Jason Wei in meters?", the gold target is "1.73 m". The predicted answer "1.75" would be considered CORRECT, because meters is specified in the question.
- For the question "What is the name of Barack Obama's wife?", the gold target is "Michelle Obama". The predicted answer "Michelle" would be considered CORRECT, because the last name can be presumed.
- Do not punish for typos in people's name if it's clearly the same name.
- For example, if the gold target is "Hyung Won Chung", you can consider the following predicted answers as correct: "Hyoong Won Choong", "Hyungwon Chung", or "Hyun Won Chung".

Grade the predicted answer of the question as one of:
A: CORRECT
B: INCORRECT
C: NOT_ATTEMPTED
Just return the letter "A", "B", or "C", with no text around it.

Figure 18: **Prompt** for grading: Part 2.

Do not generate any intermediate reasoning process. Based on the video frames, directly output a short, accurate answer to the user's question and include a confidence score (0-100) in the following JSON format:
{"answer": "Your answer here", "confidence_score": number}
Do not include any additional text or explanations outside this JSON format.

Figure 19: **Prompt** for calibration experiments.

Table 9: **Biographies of annotators** involved in `Video SimpleQA` construction.

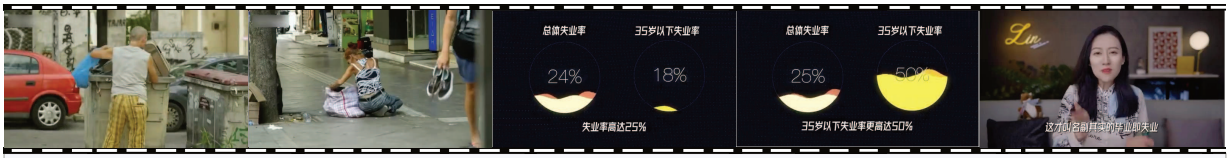
ID	Background	Language Proficiency	Avg. Time (per QA)	Agreement Rate (%)
1	CS Undergraduate	English (Fluent)	22s	92.3%
2	NLP Master	English (Native)	25s	94.1%
3	NLP PhD	English (Fluent)	23s	93.2%
4	CV PostDoc	English (Fluent)	21s	91.7%
5	Robotics Undergraduate	English (Native)	26s	89.4%
6	ML PhD	English (Fluent)	24s	92.8%
7	CS Undergraduate	English (Fluent)	28s	88.6%
8	CV PhD	English (Native)	22s	93.7%
9	Robotics Undergraduate	English (Fluent)	27s	90.1%
10	NLP PhD	English (Native)	20s	94.5%
11	ML Master	English (Fluent)	25s	91.3%
12	CS Master	English (Native)	29s	87.9%
13	CV PhD	English (Fluent)	21s	93.8%
14	Robotics Master	English (Fluent)	26s	90.7%
15	ML PhD	English (Native)	23s	92.4%
16	NLP Master	English (Fluent)	28s	89.2%
17	CS PhD	English (Fluent)	24s	91.8%
18	Robotics Undergraduate	English (Native)	30s	86.5%
19	CV Master	English (Fluent)	25s	92.1%
20	ML PhD	English (Fluent)	22s	93.6%
21	CS Undergraduate	English (Native)	27s	88.7%
22	NLP Master	English (Fluent)	26s	90.4%
23	Robotics PhD	English (Fluent)	23s	92.9%
24	CV PhD	English (Native)	24s	91.2%
25	ML PhD	English (Fluent)	21s	94.1%
26	CS Master	English (Fluent)	28s	89.8%
27	NLP Master	English (Native)	25s	92.3%
28	Robotics Undergraduate	English (Fluent)	29s	87.6%
29	CV Undergraduate	English (Fluent)	27s	90.5%
30	ML PhD	English (Native)	22s	93.4%
31	CS Master	English (Fluent)	26s	91.7%
32	NLP PhD	English (Native)	23s	93.1%
33	Robotics PhD	English (Fluent)	24s	92.6%
34	CV Undergraduate	English (Fluent)	28s	88.9%
35	ML Master	English (Native)	25s	91.4%
36	CS PhD	English (Fluent)	23s	92.8%
37	NLP Undergraduate	English (Fluent)	29s	87.3%
38	Robotics Master	English (Native)	26s	90.9%
39	CV Master	English (Fluent)	27s	89.7%
40	ML PostDoc	English (Native)	21s	94.2%

Table 10: **The video distribution** of `Video SimpleQA` benchmark (part 1).

Primary Category	Secondary Category	Tertiary Category	Count
Nature	Geology & Landscapes	Fossils	43
		Landscapes	19
		Rocks & Minerals	7
		Geomorphology	5
		Volcanic Features	7
		Coastal Landforms	6
Nature	Flora & Fauna	Animalia	18
		Marine Organisms	91
		Plantae	61
		Fungi	37
		Microorganisms	82
		Endangered Species	90
Nature	Meteorology	Weather	3
		Climate	6
		Atmospheric Phenomena	6
		Forecasting	3
Engineering	Civil & Architecture	Architecture	56
		Civil	65
		Structural Engineering	56
		Urban Planning	46
Engineering	Mechanical & Electrical	Mechanical	52
		Electrical	47
		Mechatronics	50
		Aerospace Engineering	43
Engineering	Chemical & Process	Chemical	4
		Process	3
		Biochemical Engineering	3
		Polymer Engineering	3
Engineering	Environmental & Geophysical	Environmental	22
		Geophysical	21
		Hydrology	26
		Climate Engineering	20
Science	Physical sciences	Physics	2
		Chemistry	4
		Astronomy	5
		Earth sciences	2
		Materials Science	9
		Atmospheric Science	3
		Geophysics	6
Science	Life sciences	Biology	10
		Medicine	11
		Ecology	11
		Genetics	10
		Neuroscience	7
Science	Formal sciences	Mathematics	3
		Computer Science	2
		Statistics	3
		Logic	1
Science	Applied sciences	Technology	6
		Robotics	3
		Agricultural Science	4
		Data Science	3

Table 11: **The video distribution** of `Video SimpleQA` benchmark (part 2).

Primary Category	Secondary Category	Tertiary Category	Count
Society & Culture	Arts & Recreation	Art	16
		Literature	6
		Music	30
		Entertainment	20
		Sports	12
		Dance	12
		Theatre	42
		Film	25
		Photography	10
		Games	31
Society & Culture	Beliefs & Institutions	Belief	5
		Religion	20
		Philosophy	7
		Ethics	9
		Politics	11
		Flags	8
		Government	3
		Law	3
		People	5
Society & Culture	History & Heritage	History	32
		Events	9
		Places	9
		Archaeology	24
		Heritage Sites	9
		Genealogy	9
Society & Culture	Language & Material Culture	Language	2
		Objects	9
		Food	7
		Clothing	2
		Transportation	5
		Instruments	3
		Tools	3



Question: What might be the cause of the scene in the video?
Answer: Long-term imbalance …. ☹ *[Subjective speculation]*



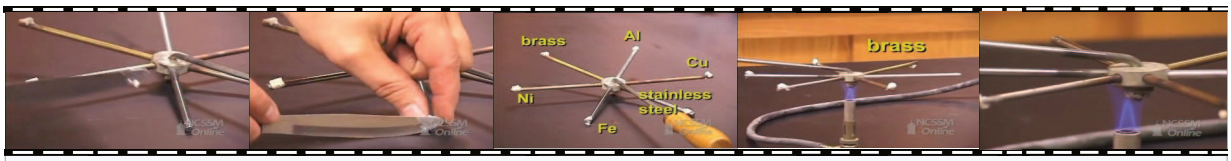
Question: What can you infer about the bird in the video?
Answer: He is potentially a healthy bird with better genes. ☹ *[Subjective speculation]*



Question: This video game seems to belong to which genre?
Answer: RPG. ☹ *[Subjective speculation]*

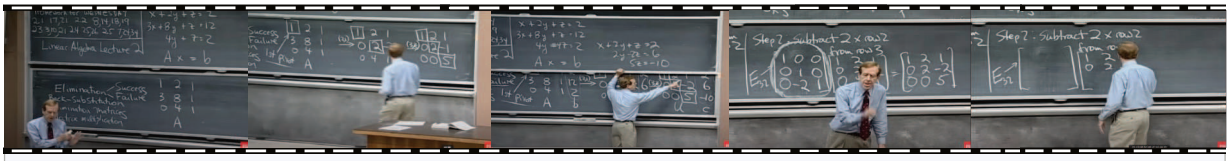


Question: What might probablly happen then?
Answer: The robot arm might try to open the bottom drawer. ☹ *[Subjective speculation]*

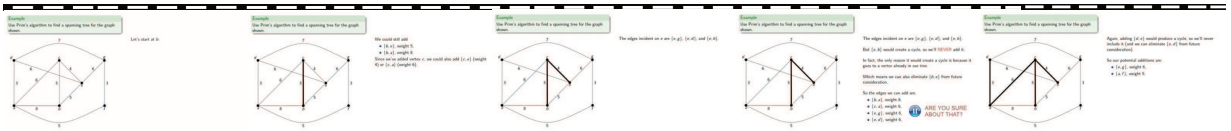


Question: According to the video, which of the following statements can be inferred?
Answer: Al has a lower specific heat than Brass ☹ *[Subjective speculation]*

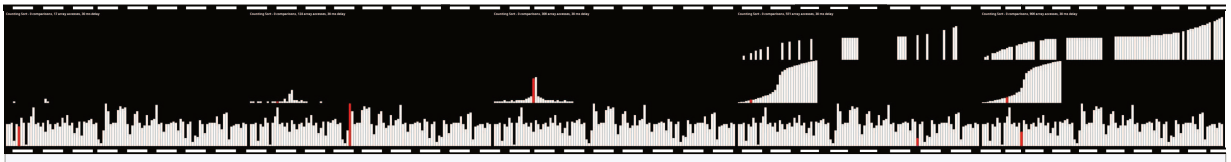
Figure 20: Examples of the inclusion of hypothetical or subjective speculation in existing discipline-based (Zhao et al. 2025; He et al. 2024a) benchmarks.



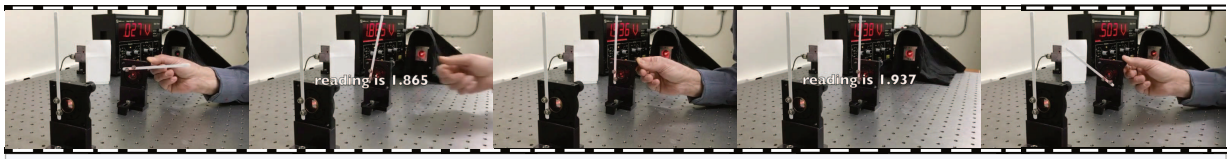
Question: Why is the (2,3) element in the last matrix -1?
Answer: ….The value should be -2. ☹ *[Entangled fact grounding and reasoning]*



Question: What is the total weight of the minimum spanning tree formed using the demonstrated algorithm?
Answer: 26 ☹ *[Entangled fact grounding and reasoning]*



Question: Given that n represents the number of elements and k denotes the range of input values, what is the worst-case time complexity of the demonstrated algorithm?
Answer: O(n + k) ☹ *[Entangled fact grounding and reasoning]*



Question: In the experiment described in the video, …. Between the two readings, how much did the number of photons reaching the photodiode change?
Answer: 3.86% ☹ *[Entangled fact grounding and reasoning]*



Question: What is his new angular velocity (omega_f) in terms of (I_0), (m), (L), and (omega_0)?
Answer: dfrac{I_0}{I_0 + 2mL^2} omega_0 ☹ *[Entangled fact grounding and reasoning]*

Figure 21: Examples of the entangled fact grounding and reasoning cases in existing discipline-based (Zhao et al. 2025; He et al. 2024a) benchmarks.



sub-question-1: What species of butterfly is shown in the video?
sub-answer-1: The Common Blue
o3-answer-1: Common blue butterfly (Polyommatus icarus)
Gemini 2.5 Pro-answer-1: Common Blue butterfly (Polyommatus icarus).
Qwen2.5-VL-72B-answer-1: Common Blue
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2010-11-09

sub-question-2: Which country honors the Common Blue as its national butterfly?
sub-answer-2: Israel
o3-answer-2: Ireland
Gemini 2.5 Pro-answer-2: Finland
Qwen2.5-VL-72B-answer-2: Sweden
reference source-2: https://en.wikipedia.org/wiki/Common_blue#Physiology

sub-question-3: Where is the capital of Israel?
sub-answer-3: Jerusalem
o3-answer-3: Jerusalem
Gemini 2.5 Pro-answer-3: Jerusalem
Qwen2.5-VL-72B-answer-3: Jerusalem
reference source-3: https://en.wikipedia.org/wiki/Israel

sub-question-4: In which year was Jerusalem declared the capital of Arabic culture?
sub-answer-4: 2009
o3-answer-4: 2009
Gemini 2.5 Pro-answer-4: 2009
Qwen2.5-VL-72B-answer-4: 2008
reference source-4: https://en.wikipedia.org/wiki/Jerusalem

multi-hop-question: In which year was the capital of the country that honors the butterfly shown in the video as its national butterfly declared the capital of Arabic culture?
multi-hop-answer: 2009
o3-answer: 2002
Gemini 2.5 Pro-answer: 2002
Qwen2.5-VL-72B-answer: 2016

Figure 22: Sampled examples in `Video SimpleQA` and the responses of typical LVLMs: part 1.



sub-question-1: What is the name of the steam tram in the video?
sub-answer-1: Feuriger Elias
o3-answer-1: Feuriger Elias
Gemini 2.5 Pro-answer-1: Feuriger Elias
Qwen2.5-VL-72B-answer-1: Feuriger Elias
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2010-12-02

sub-question-2: Which company does Feuriger Elias belong to?
sub-answer-2: Henschel & Sohn
o3-answer-2: Henschel & Sohn
Gemini 2.5 Pro-answer-2: Henschel & Sohn
Qwen2.5-VL-72B-answer-2: Henschel & Sohn
reference source-2: https://en.wikipedia.org/wiki/Tram_engine

sub-question-3: Which country is Henschel & Sohn from?
sub-answer-3: Germany
o3-answer-3: Germany
Gemini 2.5 Pro-answer-3: Germany
Qwen2.5-VL-72B-answer-3: Germany
reference source-3: https://en.wikipedia.org/wiki/Henschel_%26_Son

sub-question-4: What is the political system of Germany?
sub-answer-4: Parliamentary republic
o3-answer-4: Germany is a federal parliamentary republic
Gemini 2.5 Pro-answer-4: Parliamentary republic
Qwen2.5-VL-72B-answer-4: Parliamentary republic
reference source-4: https://en.wikipedia.org/wiki/Germany

multi-hop-question: What is the political system of the country which the company of the steam tram in the video belongs to?
multi-hop-answer: Parliamentary republic
o3-answer: Federal parliamentary republic
Gemini 2.5 Pro-answer: Parliamentary republic
Qwen2.5-VL-72B-answer: Parliamentary republic

Figure 23: Sampled examples in `Video SimpleQA` and the responses of typical LVLMs: part 2.



sub-question-1: What is the name of the movie shown in the video?
sub-answer-1: Out West
o3-answer-1: Out West ☺
Gemini 2.5 Pro-answer-1: Out West ☺
Qwen2.5-VL-72B-answer-1: The Gold Rush ☹
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2015-10-01

sub-question-2: Who is the director of the movie Out West?
sub-answer-2: Roscoe Arbuckle
o3-answer-2: Roscoe Arbuckle ☺
Gemini 2.5 Pro-answer-2: Roscoe Arbuckle ☺
Qwen2.5-VL-72B-answer-2: Buster Keaton ☹
reference source-2: https://en.wikipedia.org/wiki/Out_West_(1918_film)

sub-question-3: Which country was Roscoe Arbuckle born in?
sub-answer-3: USA
o3-answer-3: USA ☺
Gemini 2.5 Pro-answer-3: USA ☺
Qwen2.5-VL-72B-answer-3: USA ☺
reference source-3: https://en.wikipedia.org/wiki/Roscoe_Arbuckle

multi-hop-question: Which country was the director of the movie shown in the video born in?
multi-hop-answer: USA
o3-answer: USA ☺
Gemini 2.5 Pro-answer: USA ☺
Qwen2.5-VL-72B-answer: USA ☺

Figure 24: Sampled examples in `Video SimpleQA` and the responses of typical LVLMs: part 3.



sub-question-1: Which mountain is shown in the video?
sub-answer-1: Huangshan
o3-answer-1: Huangshan (Yellow Mountain) in China ☺
Gemini 2.5 Pro-answer-1: Huangshan (Yellow Mountain) in China. ☺
Qwen2.5-VL-72B-answer-1: Huangshan ☺
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2016-08-08

sub-question-2: In which province of China is Huangshan located?
sub-answer-2: Anhui
o3-answer-2: Anhui province ☺
Gemini 2.5 Pro-answer-2: Huangshan is located in the Anhui province of China. ☺
Qwen2.5-VL-72B-answer-2: Anhui ☺
reference source-2: https://en.wikipedia.org/wiki/Huangshan

sub-question-3: From the names of which two cities does the name 'Anhui' derive?
sub-answer-3: Anqing and Huizhou
o3-answer-3: Anqing and Huizhou ☺
Gemini 2.5 Pro-answer-3: Anqing and Huizhou ☺
Qwen2.5-VL-72B-answer-3: Hangzhou and Ningbo ☹
reference source-3: https://en.wikipedia.org/wiki/Anhui

multi-hop-question: From the names of which two cities does the name of the province in China, where the mountain shown in the video is located, derive?
multi-hop-answer: Anqing and Huizhou
o3-answer: Anqing and Huizhou ☺
Gemini 2.5 Pro-answer: Anqing and Huizhou ☺
Qwen2.5-VL-72B-answer: Hangzhou and Ningbo ☹

Figure 25: Sampled examples in `Video SimpleQA` and the responses of typical LVLMs: part 4.



sub-question-1: What is the title of the movie whose trailer is shown in the video?
sub-answer-1: Lawrence of Arabia
o3-answer-1: Lawrence of Arabia ☺
Gemini 2.5 Pro-answer-1: The title of the movie is Lawrence of Arabia. ☺
Qwen2.5-VL-72B-answer-1: Lawrence of Arabia ☺
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2018-10-30

sub-question-2: In which year was Lawrence of Arabia released?
sub-answer-2: 1962
o3-answer-2: 1962 ☺
Gemini 2.5 Pro-answer-2: 1962 ☺
Qwen2.5-VL-72B-answer-2: 1962 ☺
reference source-2: https://en.wikipedia.org/wiki/Lawrence_of_Arabia_(film)

multi-hop-question: Which year was the movie in the video released?
multi-hop-answer: 1962
o3-answer: 1962 ☺
Gemini 2.5 Pro-answer: 1962 ☺
Qwen2.5-VL-72B-answer: 1962 ☺



sub-question-1: What type of gymnastics movements are demonstrated in the video?
sub-answer-1: Rhythmic gymnastics
o3-answer-1: Passes through a hoop in rhythmic gymnastics ☺
Gemini 2.5 Pro-answer-1: Rhythmic gymnastics ☺
Qwen2.5-VL-72B-answer-1: Rhythmic gymnastics ☺
reference source-1: https://commons.wikimedia.org/wiki/Template:Motd/2017-07-24

sub-question-2: Which year did rhythmic gymnastics first appear as an Olympic medal event?
sub-answer-2: 1984
o3-answer-2: 1984 ☺
Gemini 2.5 Pro-answer-2: 1984 ☺
Qwen2.5-VL-72B-answer-2: 1903 ☹
reference source-2: https://en.wikipedia.org/wiki/Lawrence_of_Arabia_(film)

multi-hop-question: Which year did the type of gymnastics movements demonstrated in the video first appear as an Olympic medal event?
multi-hop-answer: 1984
o3-answer: 1984 ☺
Gemini 2.5 Pro-answer: 1984 ☺
Qwen2.5-VL-72B-answer: 1903 ☹

Figure 26: Sampled examples in `Video SimpleQA` and the responses of typical LVLMs: part 5.